



Frequency Modulators (Circuits)

Complied By:
Kartik R. Patel
EXTC Department
KJSSE, SVU

DIFFERENT TYPES OF FM.

1. NARROW BAND F.M.

We know that the bandwidth of an FM signal depends upon the frequency Deviation ($\Delta\omega = k_f \cdot x(t)$). if frequency deviation is low it means k_f is low then bandwidth will be narrow then narrow band FM is formed.

→ classification is based on modulation index $m_f = \frac{\Delta f}{f_m}$

→ $\Delta f = k_f v_m(t)$ $\Delta f \uparrow \rightarrow m_f \uparrow \rightarrow \text{No. of Sidebands} \uparrow \rightarrow B.W \uparrow$

2. WIDEBAND FM

If frequency deviation ($\Delta\omega = k_f \cdot x(t)$) is high, it means frequency sensitivity k_f is high result as bandwidth will be wide hence wide band is formed

Narrowband FM

- Narrowband FM has small bandwidth
- The modulation index m_f is small (less than 1) → AM
- Spectrum of narrowband FM consists of carrier, USB and LSB
- The maximum permissible frequency deviation is restricted to 5KHz (ΔF)

Narrowband FM (Skip)

- Mathematically narrowband FM can be expressed as

$$e_{FM}(t) = s(t) = \underbrace{E_c \sin \omega_c t}_{\text{Carrier}} + \underbrace{\frac{m_f E_c}{2} \sin(\omega_c + \omega_m) t}_{\text{USB}} - \underbrace{\frac{m_f E_c}{2} \sin(\omega_c - \omega_m) t}_{\text{LSB}}$$

$$J_0(mf) = 1,$$

$$J_1(mf) = mf/2$$

$$J_n(mf) = 0 \text{ for } n > 1$$

Narrowband FM

- Applications
 - FM Mobile communication such as
 - Police wireless
 - Ambulances
 - Taxicabs etc.

Wideband FM

- Large value of modulation index (~~mf~~) $\rightarrow m_f$
- FM wave ideally contains the carrier and an infinite number of sidebands located symmetrically around the carrier
- Such FM wave has infinite bandwidth, and called as wideband FM
- Modulation Index is higher than 1

Wideband FM

- Maximum permissible deviation is 75 KHz

- Applications

- Entertainment broadcasting applications

✓ FM Radio

✓ Television

Wideband FM (Skip)

- Mathematically wideband FM can be represented as,

$$\begin{aligned} eFM = s(t) = E_c \{ & J_0(m_f) \sin \omega_c t + J_1(m_f) [\sin(\omega_c + \omega_m) t - \sin((\omega_c - \omega_m) t)] \\ & + J_2(m_f) [\sin(\omega_c + 2\omega_m) t - \sin((\omega_c - 2\omega_m) t)] + J_3(m_f) [\sin(\omega_c + 3\omega_m) t \\ & - \sin((\omega_c - 3\omega_m) t)] + J_4(m_f) [\sin(\omega_c + 4\omega_m) t - \sin((\omega_c - 4\omega_m) t)] \dots \dots \dots (1) \end{aligned}$$

- In FM total transmitted **power** always remains **constant**, it is not dependent on modulation index (**amplitude of carrier remains constant**)

$$P_t = \frac{\left(\frac{E_c}{\sqrt{2}}\right)^2}{R} = \frac{E_c^2}{2R}$$

$$P_t = \frac{E_c^2}{2R} \text{ If } R = 1\Omega$$

Comparison of Narrowband and Wideband FM

Sr. No.	Parameter	NBFM	WBFM
1.	Modulation index	Less than or slightly greater than 1	Greater than 1
2.	Maximum deviation	5 kHz	75 kHz
3.	Range of modulating frequency	20 Hz to 3 kHz	20 Hz to 15 kHz
4.	Maximum modulation index	Slightly greater than 1	5 to 2500
5.	Bandwidth	Small approximately same as that of AM $BW = 2f_m$	Large about 15 times greater than that of NBFM. $BW = 2(\delta + f_{mmax})$
6.	Applications	FM mobile communication like police wireless, ambulance, short range ship to shore communication etc.	Entertainment broadcasting (can be used for high quality music transmission)

Basic Principle

- ✓ • Direct FM (Indirect PM)
- Direct PM (Indirect FM)
- Disadvantage of direct FM is that relatively **unstable LC oscillator** must be used to produce the carrier frequency.
- Direct FM requires addition of some **AFC** circuitry.

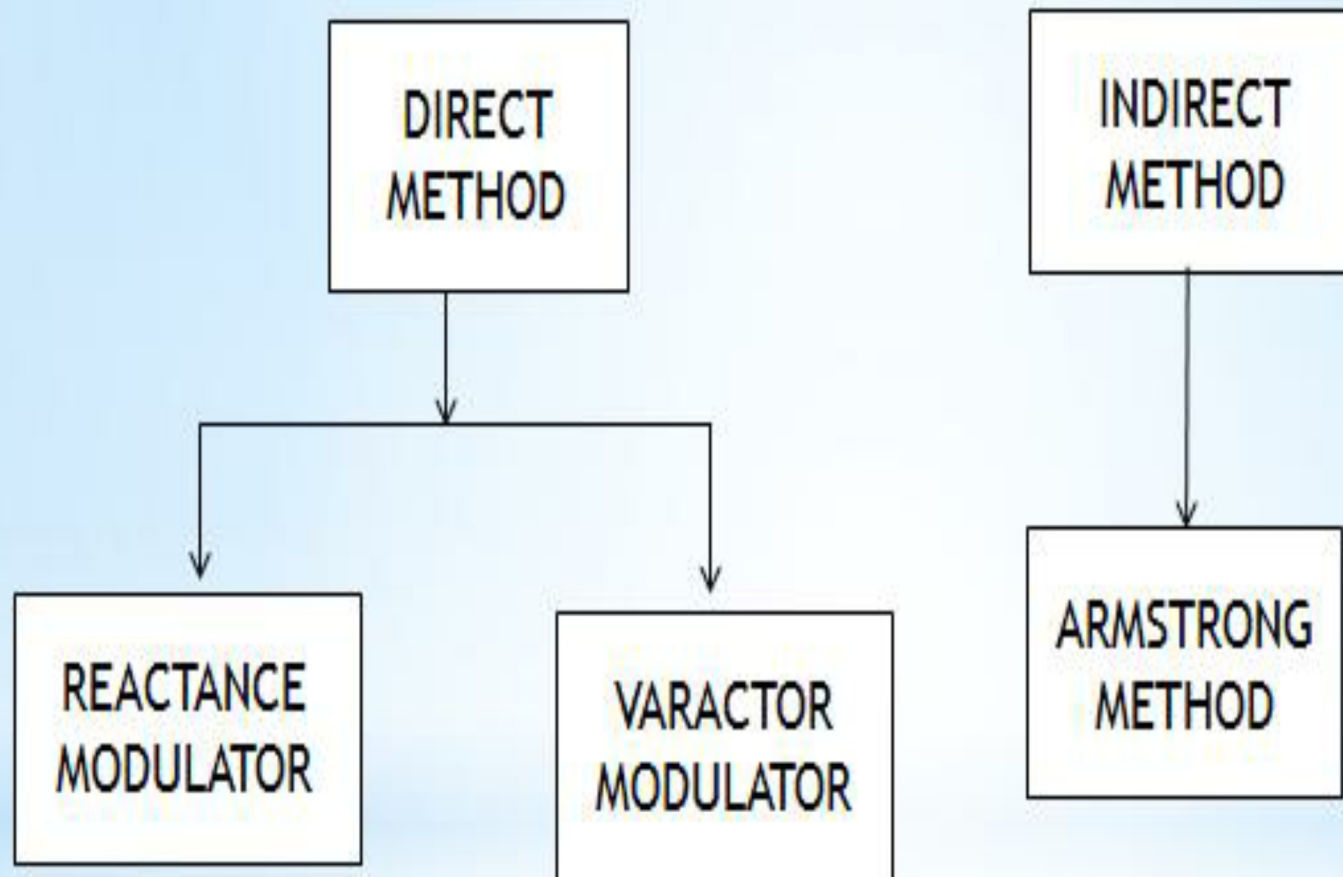
Basic Principle

- Advantage of direct FM is that high frequency deviations and modulation indices are obtained because oscillators are unstable.
- Advantage of direct PM is that the carrier oscillator is isolated from modulator circuit.
- Can be an extremely stable source such as crystal oscillator.
- High phase deviations and modulation indices are not possible because of stability of crystal oscillator.

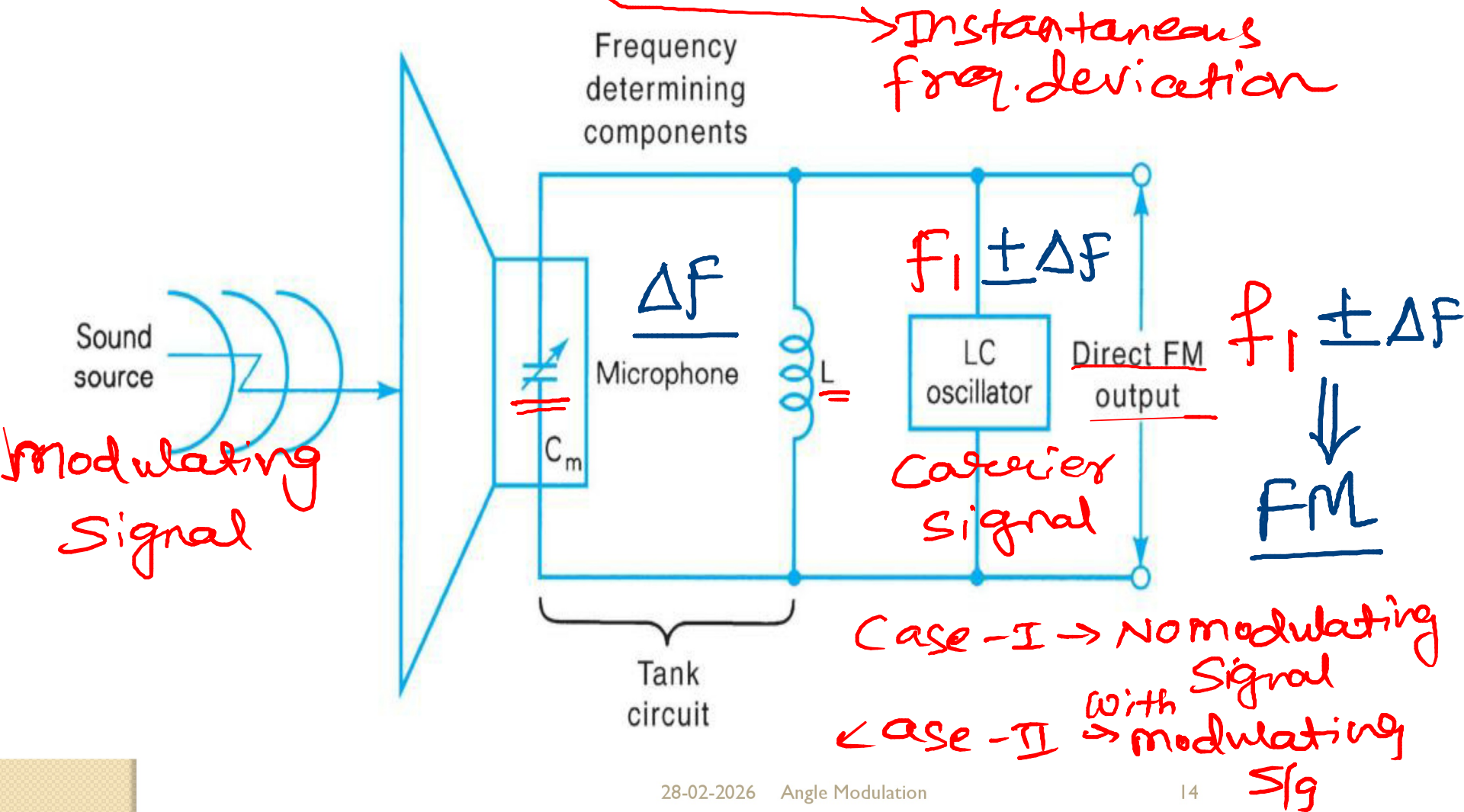
Direct FM Modulators

- Frequency of the carrier is varied directly with modulating signal.
- Instantaneous frequency deviation is directly proportional to amplitude of modulating signal.
- Varactor Diode Modulators
- FM Reactance Modulator (JFET)
- Linear-Integrated Circuit Direct FM Modulator (IC)

METHODS OF FM GENERATION



Direct FM Modulators



Indirect FM

- Direct PM Modulation Techniques

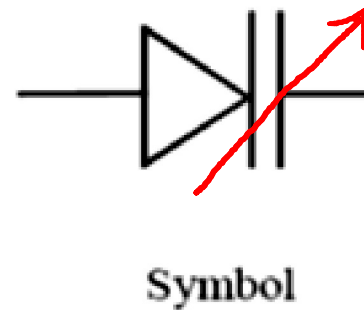
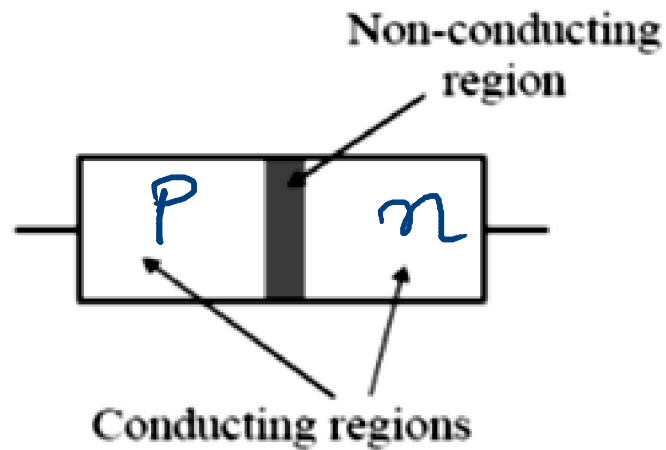
- Phase of the carrier changes according to $m(t)$
- Thus, **Indirect FM Modulation** –
- Advantages of direct PM: Uses stable crystal oscillator
- Disadvantages of direct PM: Limited phase deviation\

Direct FM

- Indirect PM Modulation Techniques

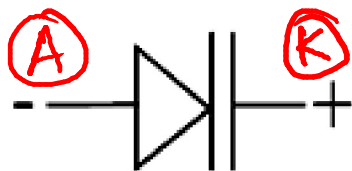
- **Direct FM Modulation** - frequency of the carrier changes according to $m(t)$
- Advantages of direct FM: easy to obtain high frequency deviation
- Disadvantages of direct FM: when using LC tanks it is not very stable, thus additional circuitry is required
- Approaches to create direct FM:
 - Varactor diode modulators
 - FM reactance modulators
 - IC-based modulators

Varactor diode



$$C = \epsilon A / d$$

Low voltage applied



Narrow non-conducting region



- more capacitance.

$$C \propto \frac{\epsilon A}{d}$$

$$X_C = \frac{1}{2\pi f C}$$

Increased voltage applied

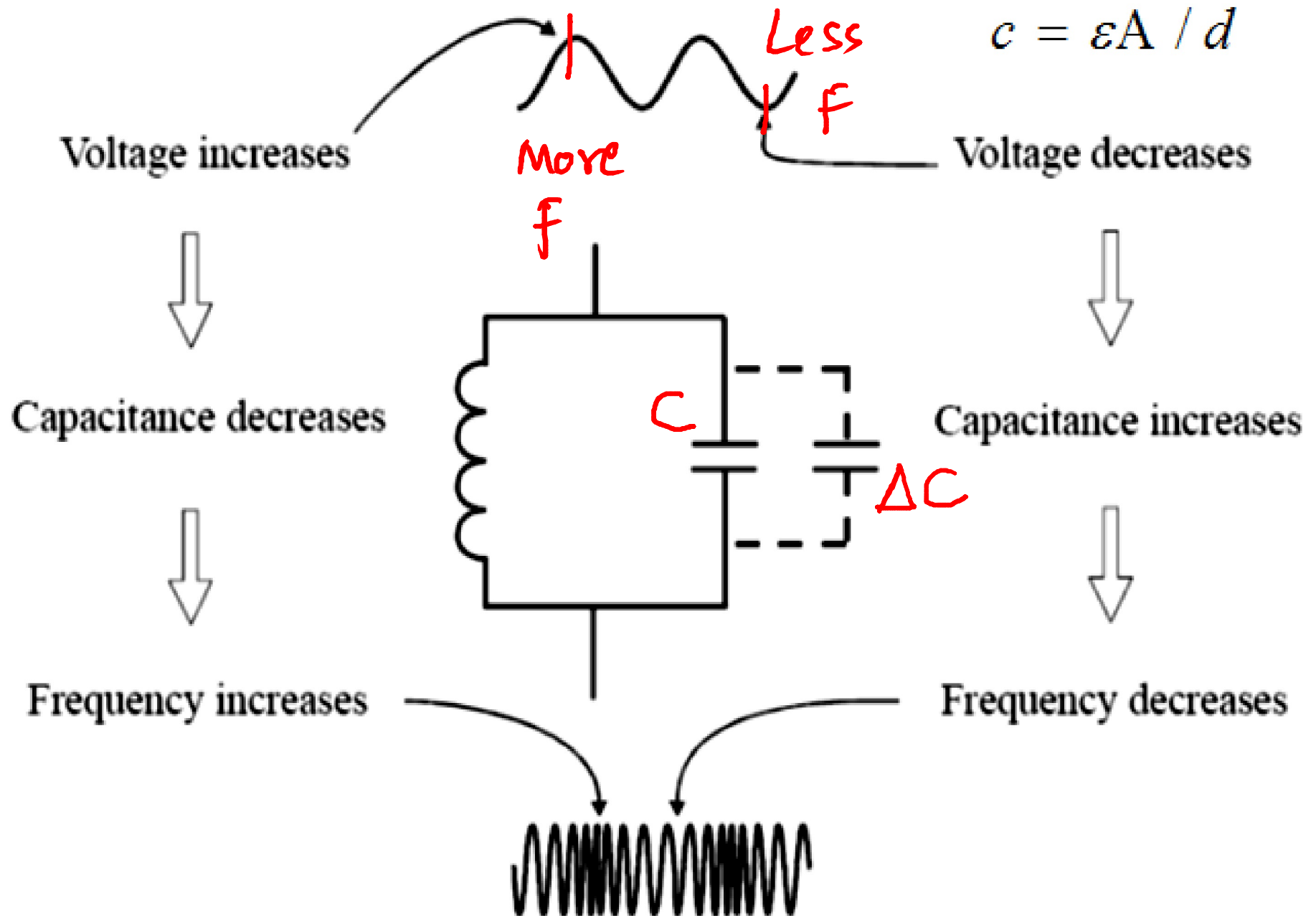


Wider non-conducting region

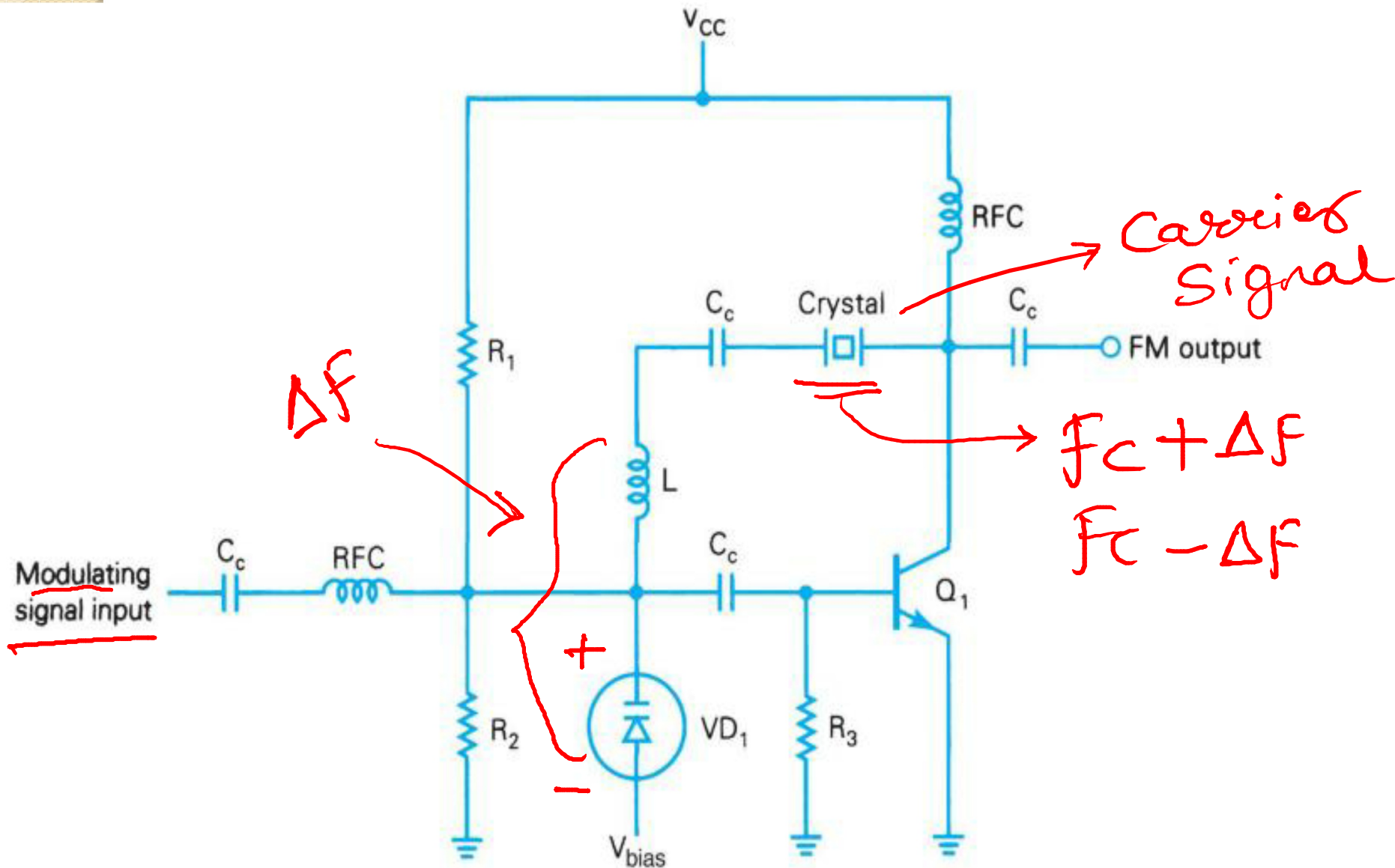


- less capacitance.

Varactor Diode FM modulator



Varactor Diode Modulators



Varactor Diode Modulators

$$f_c = \frac{1}{2\pi\sqrt{LC}} \text{ Hz}$$

$$f_c = \frac{1}{2\pi\sqrt{LC}} \text{ Hz}$$

where f_c = carrier rest frequency (hertz) $\rightarrow$ No modulation

L = inductance of the primary winding of T_1 (henries)

C = varactor diode capacitance (farads)

change in capacitance

✓ With a modulating signal applied, the frequency is

$$f = \frac{1}{2\pi\sqrt{L(C + \Delta C)}} \text{ Hz}$$

$$f = \frac{1}{2\pi\sqrt{L(C + \Delta C)}} \quad (7-53)$$

where f is the new frequency of oscillation and ΔC is the change in varactor diode capacitance due to the modulating signal. The change in frequency is

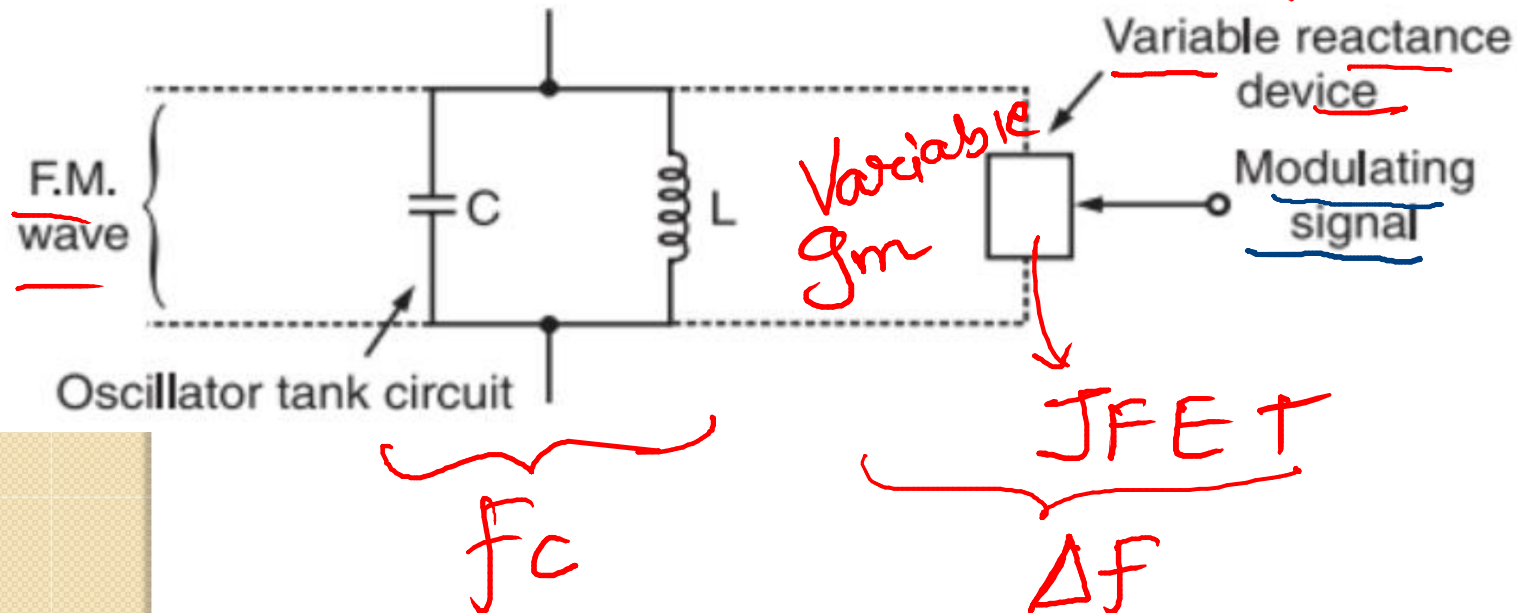
$$\Delta f = |f_c - f| \quad (7-54)$$

where Δf = peak frequency deviation (hertz)

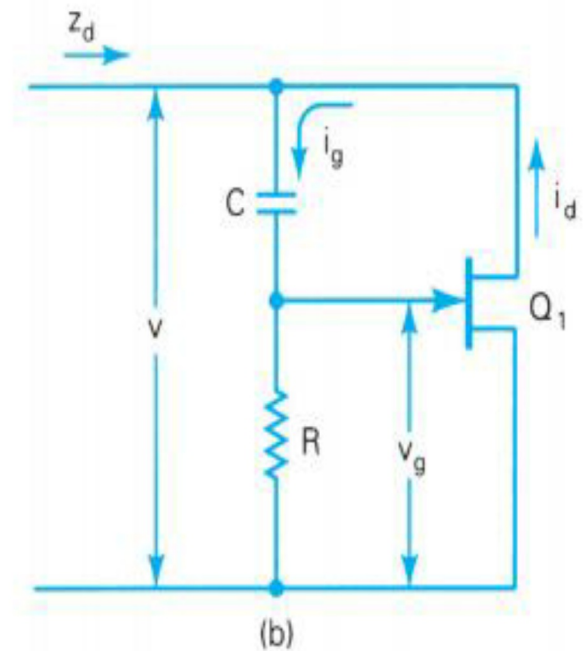
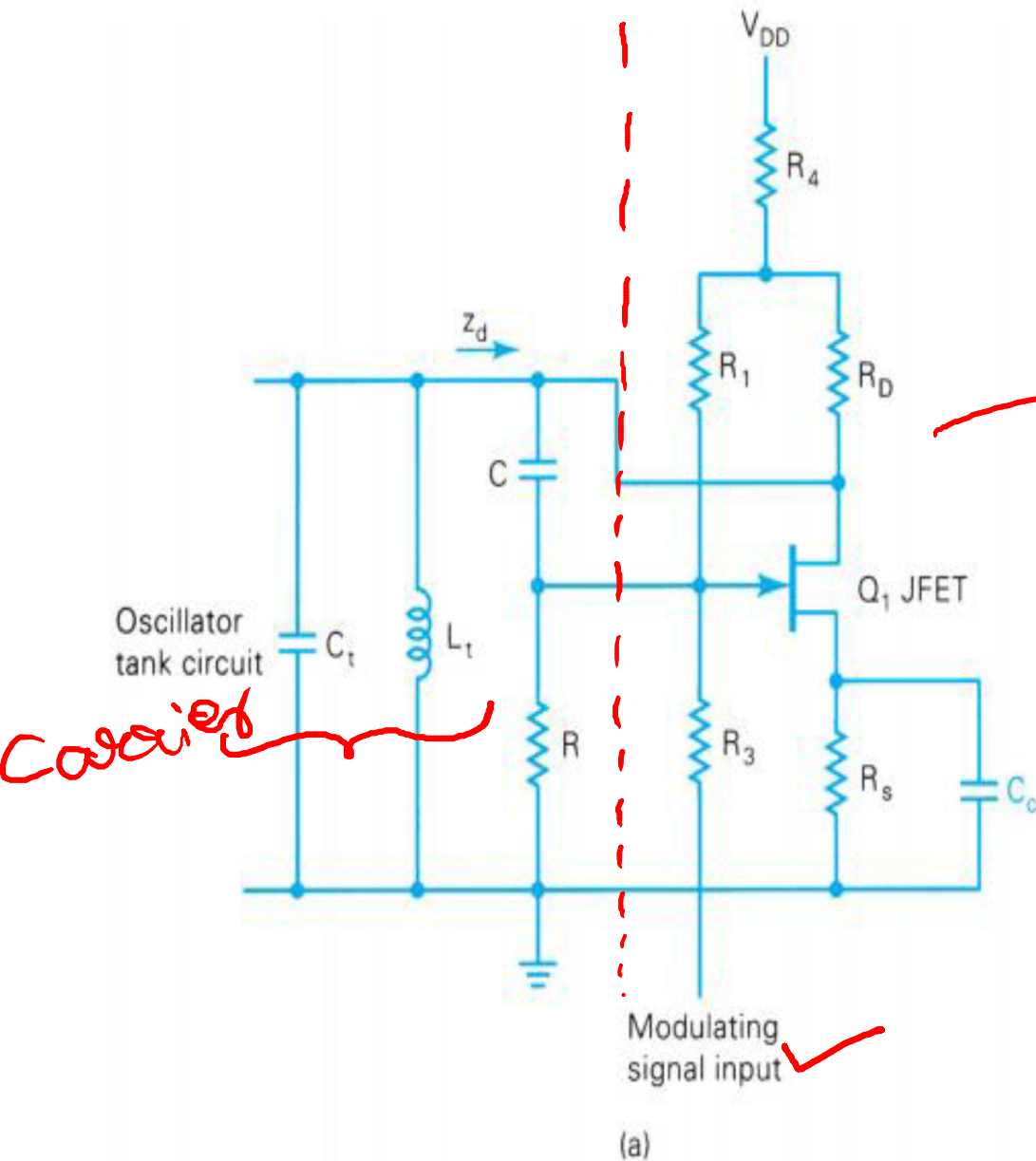
② Reactance Modulator

Capacitance + Inductance

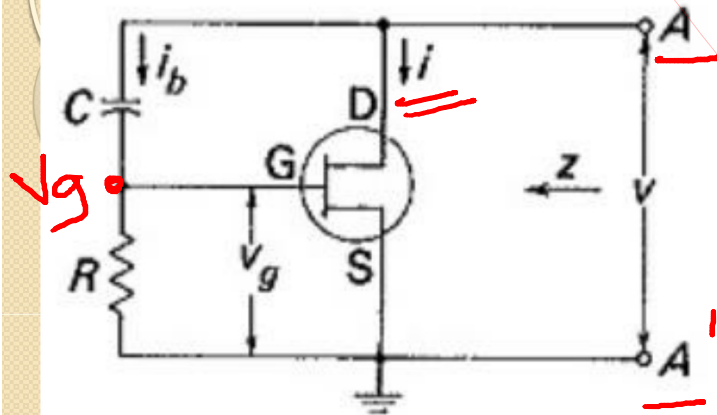
Trans conductance (g_m)



JFET Reactance Modulator



JFET Reactance Modulator



Not purely resistive

$$v_g = i_b R = \frac{Rv}{R - jX_C}$$

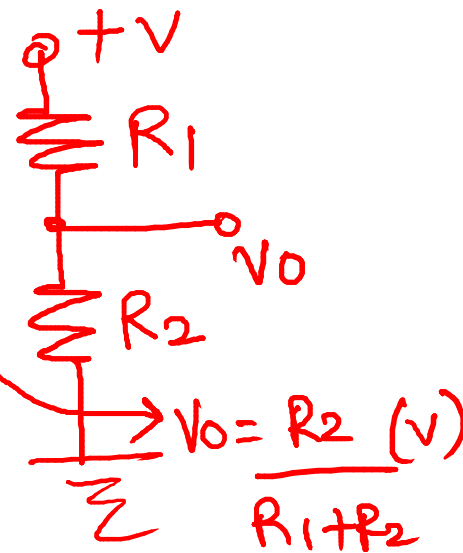
The FET drain current is

$$i = g_m v_g = \frac{g_m R v}{R - jX_C}$$

Therefore, the impedance seen at the terminals A - A' is

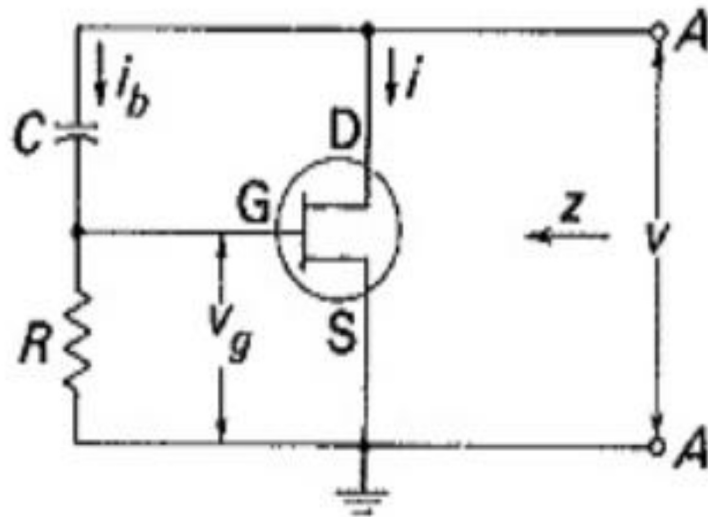
$$z = \frac{v}{i} = v \div \frac{g_m R v}{R - jX_C} = \frac{R - jX_C}{g_m R} = \frac{1}{g_m} \left(1 - \frac{jX_C}{R} \right)$$

$$\frac{g_m R v}{R - jX_C}$$



$$V_o = \frac{R_2}{R_1 + R_2} (v)$$

JFET Reactance Modulator



If $X_C \gg R$ in Equation

the equation will reduce to

$$z = -j \frac{X_C}{g_m R}$$

This impedance is quite clearly a capacitive reactance, which may be written as

$$X_{eq} = \frac{X_C}{g_m R} = \frac{1}{2\pi f g_m R C} = \frac{1}{2\pi f C_{eq}}$$

$$\frac{1}{2\pi f C}$$

From Equation it is seen that under such conditions the input impedance of the device at A – A is a pure reactance and is given by

$$C_{eq} = g_m R C$$

JFET Reactance Modulator

From Equation it is seen that under such conditions the input impedance of the device at A – A is a pure reactance and is given by

$$C_{eq} = g_m RC$$

1. This equivalent capacitance depends on the device transconductance and can therefore be varied with bias voltage.
2. The capacitance can be originally adjusted to any value, within reason, by varying the components R and C.
3. The expression $g_m RC$ has the correct dimensions of capacitance; R, measured in ohms, and g_m , measured in siemens (s), cancel each other's dimensions, leaving C as required.